A DELTA-FUNCTOR VANISHING ON SUITABLE EMBEDDINGS IS UNIVERSAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; it ends with 'We omit the verification that this defines a transformation of functors.', which is the routine functoriality check on components already constructed in the span (LANE_BRIEF 5, routine closing verification => collect), and the sentence is preserved verbatim. The corpus counts comparable proofs: STK-algebra-adjoint-tensor counts sources/stacks-algebra lines 2695-2702 containing 'We omit the verification that these constructions are mutually inverse.', and STK-algebra-loc-and-modules counts lines 1345-1353 of the same file.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Delta-functors and the universal property. Same chapter, the definitions of a delta-functor and of a universal delta-functor. Supplies the notion under discussion and the data, a transformation of functors from the degree zero part, that the construction has to extend uniquely.
- The long exact cohomology sequence of a delta-functor. Same chapter, the exactness properties of a delta-functor. Supplies the identification of the next degree of the functor with the cokernel of the map induced by the chosen embedding, which is what makes the induction step work.

Lemma 0.1. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F = (F^n, \delta_F)$ be a δ-functor from $\mathcal{A}$ to $\mathcal{B}$. Suppose that for every $n > 0$ and any $A \in \mathrm{Ob}(\mathcal{A})$ there exists an injective morphism $u : A \to B$ (depending on A and n) such that $F^n(u) : F^n(A) \to F^n(B)$ is zero. Then F is a universal δ-functor.*

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Proof. Let $G = (G^n, \delta_G)$ be a δ-functor from $\mathcal{A}$ to $\mathcal{B}$ and let $t : F^0 \to G^0$ be a morphism of functors. We have to show there exists a unique morphism of δ-functors $\{t^n\}_{n \geq 0} : F \to G$ such that $t = t^0$. We construct t^n by induction on n. For $n = 0$ we set $t^0 = t$. Suppose we have already constructed a unique sequence of transformation of functors t^i for $i \leq n$ compatible with the maps δ in degrees $\leq n$.

Let $A \in \mathrm{Ob}(\mathcal{A})$. By assumption we may choose an embedding $u : A \to B$ such that $F^{n+1}(u) = 0$. Let $C = B/u(A)$. The long exact cohomology sequence for the short exact sequence $0 \to A \to B \to C \to 0$ and the δ-functor F gives that $F^{n+1}(A) = \mathrm{Coker}(F^n(B) \to F^n(C))$ by our choice of u. Since we have already defined t^n we can set

$$t^{n+1}_A : F^{n+1}(A) \to G^{n+1}(A)$$

equal to the unique map such that

$$\begin{array}{ccc} \mathrm{Coker}(F^n(B) \to F^n(C)) & \xrightarrow{t^n} & \mathrm{Coker}(G^n(B) \to G^n(C)) \\ \downarrow{\scriptstyle \delta_{F, A \to B \to C}} & & \downarrow{\scriptstyle \delta_{G, A \to B \to C}} \\ F^{n+1}(A) & \xrightarrow{t^{n+1}_A} & G^{n+1}(A) \end{array}$$

commutes. This is clearly uniquely determined by the requirements imposed. We omit the verification that this defines a transformation of functors. $\square$